

JORDAN MANOJ CHERUVATHOOR

Machine Learning Engineer | AI Engineer | Data Scientist | Forward Deployed Engineer

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PROFESSIONAL SUMMARY

AI & Data Science Engineer (MSc Informatik, Leibniz Universität Hannover) with production experience building and deploying computer vision, NLP, and machine learning systems end-to-end. Delivered a GDPR-compliant, real-time YOLOv8 computer vision pipeline for enterprise anonymisation, automated ETL and analytics pipelines identifying €120,000+ in savings, and a peer-reviewed ensemble deep learning model (ICDAAI 2025, 94.2% accuracy). Skilled at translating ambiguous business and client requirements into deployed AI solutions across the full ML lifecycle: data engineering, model development, MLOps, and stakeholder-facing delivery. Seeking Machine Learning Engineer, AI Engineer, Data Scientist, or Forward Deployed Engineer roles at technology-driven organisations across Germany and the EU.

CORE COMPETENCIES

Machine Learning & Deep Learning: Python, PyTorch, TensorFlow, Keras, Scikit-learn, XGBoost, CNN, Ensemble Learning, Transformer Architectures, BERT, RoBERTa, Hugging Face Transformers, Hyperparameter Tuning (Optuna), Model Deployment, ONNX

Computer Vision & NLP: OpenCV, YOLOv5, YOLOv8, Real-Time Inference, Edge Inference, spaCy, NLTK, Named Entity Recognition (NER), LangChain, LangGraph, FAISS, Retrieval-Augmented Generation (RAG), Vector Indexing

Data Science & Analytics: SQL, Pandas, NumPy, Apache Spark, Power BI, Matplotlib, Seaborn, Plotly, A/B Testing, SHAP, LIME, Explainable AI (XAI), Statistical Modeling, Model Interpretability

MLOps & Engineering: Docker, Git, GitHub Actions, CI/CD, MLflow, Apache Airflow, ETL Pipelines, End-to-End ML Pipelines, FastAPI, Flask, REST APIs, Streamlit, Azure, Production-Grade Deployment

Delivery & Collaboration: Agile, Scrum, Kanban, Jira, Cross-Functional Collaboration, Client-Facing Delivery, GDPR Compliance, Stakeholder Reporting

WORK EXPERIENCE

Associate Software Engineer — Computer Vision | Cloud Orbit Technologies *Sep 2025 – Aug 2026*

- Designed and deployed a production-grade computer vision pipeline using Python, OpenCV, and YOLOv8 for automatic face and sensitive-data anonymisation across images and real-time video streams, ensuring GDPR-compliant enterprise processing.
- Developed CNN models for real-time edge inference; improved accuracy, precision, and recall through hyperparameter tuning (Optuna) and dataset augmentation, reducing false-positive rates by 15%.
- Architected end-to-end ML pipelines spanning data preprocessing, model training, validation, and containerised Docker deployment; maintained production codebase using Git and GitHub Actions in an Agile/Scrum environment with Jira.
- Collaborated cross-functionally with engineering and QA teams to validate system performance under latency constraints and lead the transition from prototype to production-ready deployment — directly engaging stakeholders on requirements and delivery timelines.

Data Science & Machine Learning Intern | White Fire Technologies *Sep 2025 – Mar 2026*

- Executed a large-scale spend analysis project on enterprise expenditure datasets (50,000+ records) using SQL and Python, deriving actionable insights for procurement optimisation and identifying €120,000+ in potential annual savings.
- Designed interactive data visualisation dashboards using Python and Power BI for C-level executive reporting, automating manual workflows and reducing report generation time by 30%.
- Built and maintained Apache Airflow ETL pipelines for automated data ingestion and transformation; implemented CI/CD workflows using GitHub Actions for version control and automated testing.
- Followed Agile/Scrum ticketing workflows (Jira) for sprint tracking and iterative delivery across a distributed UK–India cross-functional team.

AI & Data Science Researcher | Independent Research — ICDAAI 2025 Publication *Dec 2024 – Mar 2025*

- Authored and published peer-reviewed paper: “Advanced Ensemble Learning and Feature Enhancement for Robust Breast Cancer Classification in Histopathological Images,” ICDAAI 2025 (Paper ID: F-ICDAAI-V3-25-1026).
- Developed an ensemble deep learning model combining CNNs and feature enhancement techniques, achieving 94.2% classification accuracy on the BreakHis and PatchCamelyon datasets using PyTorch and TensorFlow/Keras.
- Applied SHAP and LIME explainable AI (XAI) frameworks to interpret model predictions and improve clinical interpretability for safe AI deployment in EU healthcare contexts.

EDUCATION

MSc Informatik (Computer Science) | Leibniz Universität Hannover, Germany

Apr 2026 (Ongoing)

Focus areas: Machine Learning, Deep Learning, Data Science, Software Engineering, Intelligent Systems, Distributed Systems

B.Tech — Artificial Intelligence & Data Science | Karunya Institute of Technology and Science, India *Aug 2021 – May 2025*
Distinction Awarded. Core modules: Neural Networks, NLP, Machine Learning, Database Systems, Data Encryption, Agile Development, Software Architecture

SELECTED PROJECTS

Sentiment-Insight | Python, Hugging Face, RoBERTa, PyTorch, ONNX

- Fine-tuned RoBERTa transformer for automated sentiment analysis on 15,000+ product and social media reviews, achieving 92.3% F1-score across five classes; built full NLP pipeline with ONNX export for low-latency production deployment.

Deep-Research-Bot | Python, LangChain, LangGraph, FAISS, Gemini API, Docker

- Built an autonomous AI research system with web crawling, FAISS vector indexing, and a multi-agent Retrieval-Augmented Generation (RAG) pipeline; deployed as a containerised Streamlit app with sub-2-second retrieval latency across 10,000+ documents.

EntityNet — Visualizing Textual Relationships | Python, spaCy, NetworkX, NER

- Developed an NLP system for entity relationship extraction from unstructured text using spaCy NER and dependency parsing (89% accuracy); built an interactive knowledge graph with NetworkX for investigative analytics.

Diabetic Readmission Risk Predictor | Python, Scikit-learn, Streamlit, SHAP

- Developed an ML model predicting 30-day diabetic patient readmission with 87% AUC-ROC; deployed as an interactive Streamlit app with real-time risk scoring and SHAP-based explainability for clinical decision support.

Breast Cancer Classification — Histopathology Images | Python, PyTorch, TensorFlow, OpenCV

- Developed a dual-stream high-order deep CNN (DsHoNet) for automated breast cancer classification across four computer vision tasks on BreakHis and PatchCamelyon; applied ensemble learning, Optuna hyperparameter optimisation, and SHAP/LIME explainability.

RESEARCH PUBLICATIONS

Jordan Manoj Cheruvathoor. “Advanced Ensemble Learning and Feature Enhancement for Robust Breast Cancer Classification in Histopathological Images.” *Proceedings of ICDSAAI 2025*. Paper ID: F-ICDSAAI-V3-25-1026.

CERTIFICATIONS

- Google: Advanced Data Analytics (6 certificates + Capstone), Foundations of UX Design
- Microsoft: Power BI Data Analyst Professional Certificate, Azure Fundamentals, Excel for Data Analysis
- IBM: Enterprise-Grade AI, Tools for Data Science, Data Science Methodology
- Cisco: Cloud Security, Introduction to Packet Tracer
- Other: TensorFlow Developer Certificate, Deep Learning Fundamentals, PCAP Python Essentials (OpenEDG), SQL for Data Analysis (Internshala)

LANGUAGES

English: C1 Proficient (fluent in technical and business communication) **German:** A2 Basic (actively progressing toward B1)
Work Authorization: Indian Nationality; German Student Visa, full-time eligible post-graduation; open to EU relocation.